

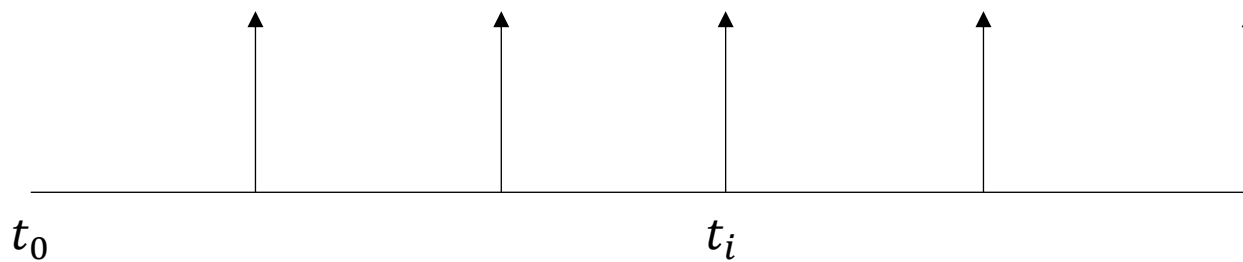
RFR Regime

(4. SOFR Swaps – Valuation & Sensitivities)

IRS Pricing - Annuity

- ▶ Given a stream of coupon payments, Annuity (A) is defined as the present value of unit coupons on unit Notional

$$A = \sum_{i=1}^n \tau_i P(t_0, t_i)$$



- ▶ We will enhance this definition to include a notional amount N

$$A_N = N \sum_{i=1}^n \tau_i P(t_0, t_i)$$

(when Notional is constant)

$$A_{N_i} = \sum_{i=1}^n N_i \tau_i P(t_0, t_i)$$

(when Notional is variable)

IRS Pricing – Value of Fixed/Floating Legs

► Value of the Fixed Leg

$$PV^{\text{Fixed Leg}} = r^{\text{Fixed}} \sum_{i=1}^n N_i \tau_i P(t_0, t_i) = r^{\text{Fixed}} \cdot A_{N_i}^{\text{Fixed}}$$

► Value of the Floating Leg

$$PV^{\text{Floating Leg}} = \sum_{j=1}^m N_j (r_j + s) \tau_j P(t_0, t_j)$$

Sometimes, it's helpful to bifurcate the floating and spread components

$$PV^{\text{Floating Leg}} = \underbrace{\sum_{j=1}^m N_j r_j \tau_j P(t_0, t_j)}_{\text{(floating)}} + \underbrace{s \sum_{j=1}^m N_j \tau_j P(t_0, t_j)}_{\text{(fixed)}} = \sum_{j=1}^m N_j r_j \tau_j P(t_0, t_j) + s A_{N_j}^{\text{Float}}$$

IRS Pricing – Par Rate

- ▶ The Fixed rate that makes the NPV of swap = 0 is called as the **par rate**. At inception typically swaps have 0 values and the fixed rate = par rate. Swaps that are executed with fixed rate = par rate are called **par swaps**.
- ▶ To find par rate, we make fixed leg = floating leg (without spread)

$$Nr^{\text{Fixed}}A^{\text{Fixed}} = N \sum_{j=1}^m r_j \tau_j P(t_0, t_j)$$

$$p = \frac{\sum_{j=1}^m r_j \tau_j P(t_0, t_j)}{A^{\text{Fixed}}}$$

$$p = \frac{PV_{s=0}^{\text{Floating Leg}}}{A_N^{\text{Fixed}}} = \frac{PV^{\text{Floating Leg}} - sA_N^{\text{Float}}}{A_N^{\text{Fixed}}}$$

Receiver Swap and Payer Swap

- ▶ A receiver swap is the one that receives fixed and pays floating. Similarly, a payer swap is the one that pays fixed and receives floating. Let's define indicator ϕ to denote the position, so that

$$\phi = \begin{cases} 1, & \text{Receiver} \\ -1, & \text{Payer} \end{cases}$$

- ▶ Value of swap is given as

$$PV^{\text{swap}} = \phi(PV^{\text{Fixed Leg}} - PV^{\text{Floating Leg}})$$

$$PV^{\text{swap}} = \phi \left(r^{\text{Fixed}} A_N^{\text{Fixed}} - \sum_{j=1}^m N(r_j + s) \tau_j P(t_0, t_j) \right)$$

$$PV^{\text{swap}} = \phi[(r^{\text{Fixed}} - p)A_N^{\text{Fixed}} - sA_N^{\text{Float}}]$$

IRS Risk – Macaulay Duration

- **Macaulay Duration** : While Macaulay Duration is classically understood in the context of a bond, it's common market practice to calculate Duration for Fixed and Floating leg separately. There is no concept of a common yield for the swap.

For bonds we have

$$D_{Mac} = \sum_{i=1}^n t_i \left(\frac{\text{PV of } i\text{th Bond Coupon}}{\text{Bond Price}} \right)$$

For fixed-leg of the swap, we have

$$D_{Mac}^{\text{Fixed}} = \sum_{i=1}^n t_i \left(\frac{\text{PV of } i\text{th Fixed Coupon}}{\text{PV of Fixed Leg}} \right) = \frac{r^{\text{Fixed}} \sum_{i=1}^n t_i N_i \tau_i P(t_0, t_i)}{r^{\text{Fixed}} \sum_{i=1}^n N_i \tau_i P(t_0, t_i)}$$

$$D_{Mac}^{\text{Fixed}} = \left(\frac{\sum_{i=1}^n t_i N_i \tau_i P(t_0, t_i)}{A_{N_i}^{\text{Fixed}}} \right)$$

IRS Risk – Macaulay Duration

- ▶ Similarly for floating-leg, we can calculate Macaulay Duration as

$$D_{Mac}^{Fixed} = \sum_{j=1}^m t_j \left(\frac{\text{PV of } j\text{th Floating Coupon}}{\text{PV of Floating Leg}} \right) = \frac{\sum_{j=1}^m t_j N_j r_j \tau_j P(t_0, t_j)}{\sum_{j=1}^m N_j r_j \tau_j P(t_0, t_j)}$$

- ▶ Spread duration can be calculated separately as it behaves like a fixed leg

$$D_{Mac}^{spread} = \left(\frac{\sum_{j=1}^m t_j N_j \tau_j P(t_0, t_i)}{A_{N_j}^{Spread}} \right)$$

IRS Risk – Modified Duration

- ▶ Modified duration in case of a bond refers to the % change in price because of change in yield. It's given as

$$D_{mod} = \frac{D_{Mac}}{(1 + y)}$$

- ▶ But there is no such concept as a yield for a swap. However, when bond's coupon rate matches its yield, bond trades at par. Similarly, when a swap's fixed coupon rate = par rate, the swap trades at par (value = 0). So, analogously we can treat the swap **par rate as the yield**.

$$D_{mod} = \frac{D_{Mac}}{(1 + p)}$$

IRS Risk – PV01



- ▶ In general, PV01 (present value of basis point) represents the sensitivity to 1 bp change in interest rates. In the context of swap, this refers to the sensitivity of the fixed leg to 1 bp increase in fixed rates. This metric ignores changes in discount factors

$$PV01^{\text{swap}} = (\phi A_{N_i}^{\text{Fixed}}) / 10000$$

While this is not accurate measure of sensitivity to yield curve, it's simple and fast to calculate analytically. A related concept is DV01 (next slide) which is more accurate. For par swaps PV01 is equal to DV01.

IRS Risk – DV01

- ▶ DV01 (dollar value of basis point) represents the sensitivity of the swap to 1 bp downward-shift in par rates. This includes changes in both forecast curve and discount curve making it a more comprehensive measure. Mathematically

$$DV01^{\text{swap}} = - \left(\frac{dPV^{\text{swap}}}{dp} \right) / 10000$$

Since

$$PV^{\text{swap}} = \phi \left[(r^{\text{Fixed}} - p) A_N^{\text{Fixed}} - s A_N^{\text{Float}} \right]$$

$$\left(\frac{dPV^{\text{swap}}}{dp} \right) = -\phi \left(A_N^{\text{Fixed}} - (r^{\text{Fixed}} - p) \frac{dA_N^{\text{Fixed}}}{dp} - s \frac{dA_N^{\text{Float}}}{dp} \right)$$

After simplification, one obtains

$$DV01^{\text{swap}} = \phi (A_N^{\text{Fixed}} - (r^{\text{Fixed}} - p) A_N^{\text{Fixed}} D_{\text{Mod}}^{\text{Fixed}} - s A_N^{\text{Float}} D_{\text{Mod}}^{\text{Float}}) / 10000$$